Differentiation on a Computer I

Finite differences and forward-mode automatic differentiation

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How PyTorch checks analytical gradients

The bug that does not crash

You extend PyTorch with a custom operation — a fused kernel, a new layer. Autograd asks **you** to supply its derivative.

Get it slightly wrong, and **nothing crashes**:

- the loss still goes down (mostly) — wrong gradients are often “downhill-ish”
- the model trains, ships… and is quietly worse than it should be

A wrong gradient may produce no exception or NaN. A numerical gradient check gives an independent comparison before the operation is used in training.

A numerical check: `torch.autograd.gradcheck`

Every serious framework ships a gradient checker. PyTorch's:

```python
import torch
from torch.autograd import gradcheck

def my_op(x):                       # your shiny new operation
    return torch.sin(x) * x**2

x = torch.randn(4, dtype=torch.float64, requires_grad=True)
print(gradcheck(my_op, (x,)))       # True — backward() is correct
```

And what does it compare `backward()` against? **This**:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$ — a slope, measured with a tiny nudge

Three design questions in that one call

PyTorch compares analytical gradients with finite-difference secants. Three implementation choices need explanation:

1. **Accuracy** — the numerical comparison is an approximation. How large can its error be?
2. **Numerical choices** — the default tolerances are designed for float64, `eps = 1e-6`, and the formula is symmetric. We will derive why.
3. **Cost** — if nudging is useful for checking, why is it not used to train?

All three fall out of one lecture. The answers involve L2′s machine epsilon, L7′s Taylor remainder — and a number system where $\varepsilon^2 = 0$.

Two ways to compute a derivative

Finite differences approximate a derivative and incur truncation and rounding error. Automatic differentiation applies the chain rule to the executed program, without choosing a step size.

Two parts:

- **Finite differences**: truncation decreases with h, while rounding error increases. Their sum produces a U-shaped error curve and an optimal scale for h ⭐.
- **Forward-mode AD**: dual numbers propagate a value and tangent through each primitive, so there is no finite-difference truncation or cancellation.

Learning outcomes

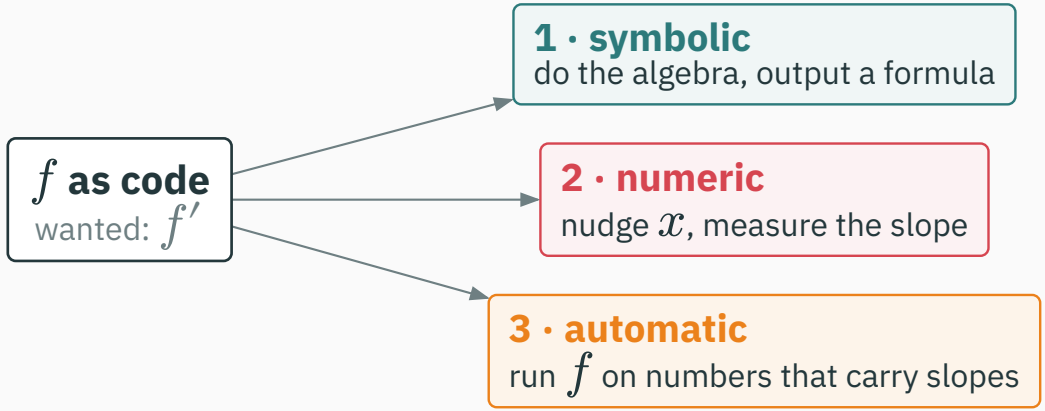
By the end of this lecture you will be able to:

1. Estimate derivatives with **forward and central differences** and predict their error orders.
2. Explain the tradeoff between **truncation and rounding error**, and read an error-vs-h **U-curve**.
3. Derive the **optimal step** $h^* \approx \sqrt{\varepsilon}$ and the "half your digits" floor. ⭐
4. Explain why finite differences **cannot scale** to 10^6 parameters.
5. Compute with **dual numbers** $a + b\varepsilon$, $\varepsilon^2 = 0$ — and show that they apply derivative rules exactly to the represented program (up to ordinary floating-point rounding).
6. Run **forward-mode autodiff** by hand: a (value, tangent) trace, one pass per input.

Three approaches to a derivative

Symbolic, numerical, and automatic differentiation

V VISUAL



This trichotomy — **symbolic / numeric / automatic** — organizes everything about differentiation on machines (it's the framing of the Baydin et al. survey in your reading).

The first two methods follow directly from calculus. Automatic differentiation will be derived from computation graphs in L11.

Symbolic differentiation: do the algebra

What you did in L7, mechanized (SymPy, Mathematica): apply the rules, output a **formula**.

$$f(x) = x^2 \sin(x) \quad \rightarrow \quad f'(x) = 2x \sin(x) + x^2 \cos(x) \quad \text{— exact } \checkmark$$

Exact, human-readable… and it needs f to **be** a formula. Your training loss — Python loops, branches, lookups — has no closed form to hand over.

And formulas have a darker problem. Watch what happens when we **compose** — the one thing deep learning does all day.

Symbolic differentiation has a swelling problem

Iterate a humble map (composition = depth, exactly like stacked layers): $\ell_1 = x$, then $\ell_{k+1} = 4\ell_k(1-\ell_k)$:

Depth	The formula
ℓ_2	$4x(1-x)$
ℓ_3	$16x(1-x)(1-2x)^2$
ℓ_4	$64x(1-x)(1-2x)^2(1-8x+8x^2)^2$

Now differentiate ℓ_4 — four compositions deep — and expand:

$$\mathrm{d}\ell_4/\,\mathrm{d}x = -131072x^7 + 458752x^6 - 638976x^5 + 450560x^4 - 168960x^3 + 32256x^2 - 2688x + 64$$

Expression swell: each level of composition multiplies the derivative's size. A 100-layer network's symbolic gradient would be astronomical.

Numerical differentiation: measure a secant slope

No algebra at all — **nudge and divide**:

```
def derivative(f, x, h=1e-8):
    return (f(x + h) - f(x)) / h    # works on ANY f — even a black box
```

- Three lines. No formula needed — loops, branches, whole programs welcome.
- The result is approximate. Its error depends on both h and the floating-point format.

The default `h=1e-8` is not universally optimal. The ⭐ derivation will show how a useful scale follows from curvature and machine precision.

Automatic differentiation: propagate derivatives

Automatic differentiation (AD): run the **program** of f, but on values that carry their own derivatives — every $+$ and $*$ updates both.

- No finite-difference approximation: it differentiates the sequence of implemented primitives; floating-point evaluation still rounds normally
- **Formula-free** like numeric — works on code, never builds an expression
- Cost: a small constant times the cost of running f itself

PyTorch's `autograd` is one implementation. Today we build forward mode by hand; L11 develops reverse mode and backpropagation.

First we quantify the error and cost of finite differences.

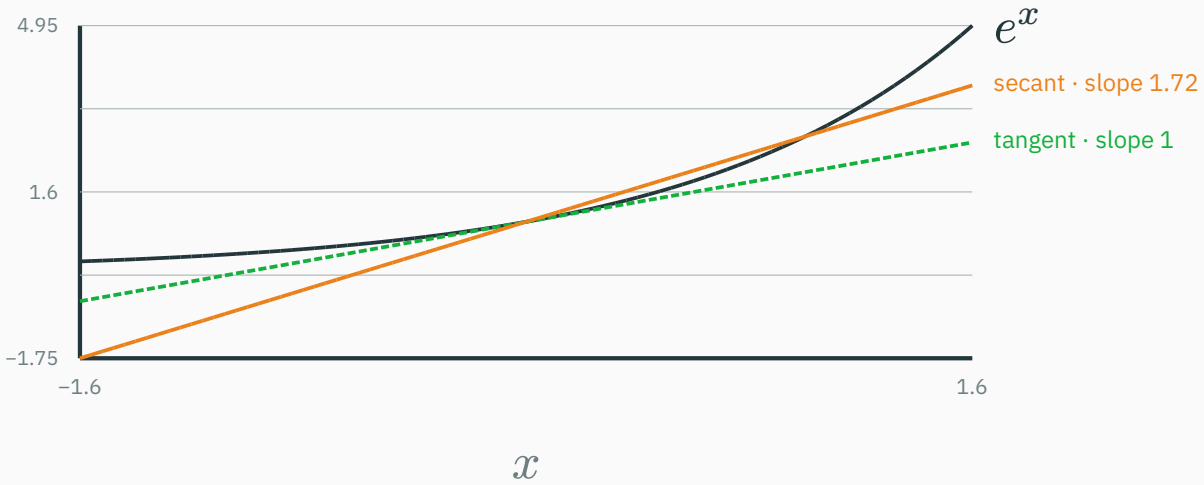
Forward differences: two sources of error

L7′s limit, with the limit amputated

V VISUAL

L7 *defined* the derivative as a limit of secant slopes. A computer can’t take limits — it can only **stop early**:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \qquad \Rightarrow \qquad D_h f(x) := \frac{f(x+h) - f(x)}{h}$$



$f = e^x$ at $x = 0$, true slope 1: the secant with $h = 1$ overshoots to 1.72. Shrink h…

First contact: it kind of works

Test on $f = e^x$ at $x = 0$, where the truth is exactly $f'(0) = e^0 = 1$:

h	$D_h f(0)$ **(float64)**	**error**
10^{-1}	1.0517091808…	5.2×10^{-2}
10^{-2}	1.0050167084…	5.0×10^{-3}
10^{-3}	1.0005001667…	5.0×10^{-4}

Divide h by 10 → error divides by 10. Extrapolate the pattern: $h = 10^{-16}$ should give **16 correct digits**.

And look closer: the error isn't just *order* h — it is almost exactly $h/2$. That "2" is not a coincidence. Taylor knows why.

Truncation error from Taylor's theorem

L7 callback — Taylor's remainder form is an *equality* (some ξ between x and $x+h$): $f(x+h) = f(x) + hf'(x) + (h^2/2)f''(\xi)$. Rearrange the forward difference:

$$D_h f(x) = f'(x) + \underbrace{\frac{h}{2} f''(\xi)}_{\text{truncation error} \;=\; O(h)}$$

- For e^x at 0: $f'' \approx 1$ → error $\approx h/2$ — exactly the table's 5×10^{-3} at $h = 10^{-2}$ ✓
- Truncation $\propto h$: you stopped Taylor early and pay the curvature you ignored. Cure: **shrink h**.

So drive $h \to 10^{-16}$ and collect 16 digits… right? You spent L2 learning why not.

Subtraction cancels leading digits

L2 callback: catastrophic cancellation — subtracting near-equals promotes rounding garbage to the front. And $f(x+h) - f(x)$ is a subtraction of near-equals **by design**.

Take $h = 10^{-8}$. What float64 actually stores and computes:

```
e^h stored   =  1.00000000999999993922529O3…  ← last digits: rounding noise
1.0          =  1.00000000000000000000000000
difference   =  0.00000000999999993922529O3…  ← 8 leading digits annihilated
```

Both inputs carried ~16 good digits. Their difference carries **~8** — then `/ h` scales the surviving noise up by 10^8.

Started with 16 digits. Kept 8. The estimate: $0.9999999939\ldots$ **— noise wearing a derivative costume.**

A rounding-error model grows as ε/h

L2′s rounding rule: each stored value is off by a relative δ, $|\delta| \le \varepsilon$ — recall $\varepsilon_{64} \approx 2.2 \times 10^{-16}$, $\varepsilon_{32} \approx 1.2 \times 10^{-7}$.

So each evaluation of f is really $f \cdot (1+\delta)$ — wrong by up to $\varepsilon\ |f|$. The numerator inherits up to $2\varepsilon\ |f|$ of garbage, and dividing by h **amplifies** it:

$$\text{rounding error of } D_h f \approx \frac{2\varepsilon\ |f(x)|}{h} \quad \text{— grows as } h \text{ shrinks!}$$

- Truncation shrinks with h; rounding error grows with $1/h$.

For large h, truncation dominates. For small h, rounding dominates. Between them is a minimum-error scale.

Measured error across step sizes

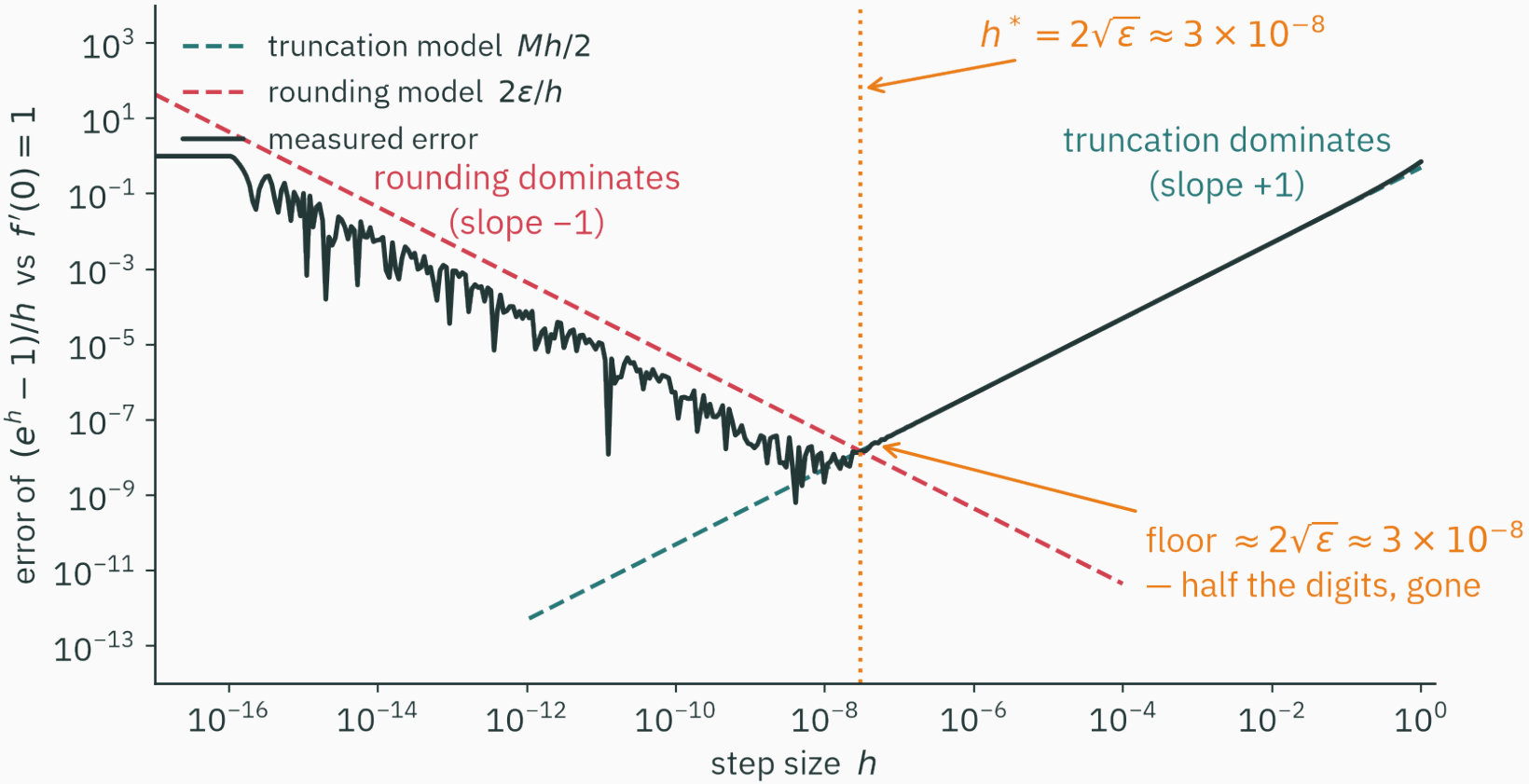
$f = e^x$, $x = 0$, float64 — every digit below is real:

h	$D_h f(0)$	**error**
10^{-4}	1.0000500017…	5 × 10⁻⁵
10^{-8}	**0.9999999939…**	**6 × 10⁻⁹** — the best it ever gets
10^{-12}	1.0000889006…	9 × 10⁻⁵ — worse again!
10^{-15}	1.1102230246…	1 × 10⁻¹ — that's ε_{64}'s face
10^{-16}	**0.0000000000**	**100%**

The last row is pure L2: $1 + 10^{-16}$ **rounds to 1** — the nudge fell into the float gap, the numerator is exactly 0, and the estimator reports that the derivative of *everything* is zero. With total confidence.

The U-curve

V VISUAL



Error vs h for $D_h e^x$ at $x = 0$ in float64: truncation dominates on the right and rounding on the left.

How to read a U-curve

- **Right wall, slope $+1$**: truncation, $\propto h$ — smooth, predictable, Taylor's receipt
- **Left wall, slope -1**: rounding, $\propto \varepsilon/h$ — jagged, because rounding noise is erratic, not smooth
- **The floor** near $h \approx 10^{-8}$: error $\approx 10^{-8}$ — the best this method will *ever* do in float64

There is no winning h — only a least-bad one. Every finite-difference scheme has a floor it cannot go below.

We now derive the location and scale of the minimum.

⭐ The best possible step

⭐ Step 1: model the total error

D DERIVATION

Total error = truncation + rounding. Bound $|f''| \leq M$ near x (curvature scale, L7), take $|f(x)| \approx 1$ for clean bookkeeping (our e^0 case exactly):

$$E(h) = \underbrace{\frac{M}{2}h}_{\text{truncation — grows with } h} + \underbrace{\frac{2\varepsilon}{h}}_{\text{rounding — grows with } 1/h}$$

- One term climbs, one falls: $E(h)$ is U-shaped — we **modeled** the measured curve
- Those are the two dashed lines drawn on the U-curve figure — this is a model you can check

Minimizing a function of one variable… we happen to own a lecture on that (L7).

⭐ Step 2: minimize — with L7′s own calculus

D DERIVATION

Set the derivative of the error (the derivative of the error of a derivative!) to zero:

$$E'(h) = \frac{M}{2} - \frac{2\varepsilon}{h^2} = 0 \quad \Rightarrow \quad h^* = 2\sqrt{\frac{\varepsilon}{M}}$$

At the optimum, plug h^* back in — the two modeled contributions are equal:

$$E(h^*) = \sqrt{\varepsilon M} + \sqrt{\varepsilon M} = 2\sqrt{\varepsilon M}$$

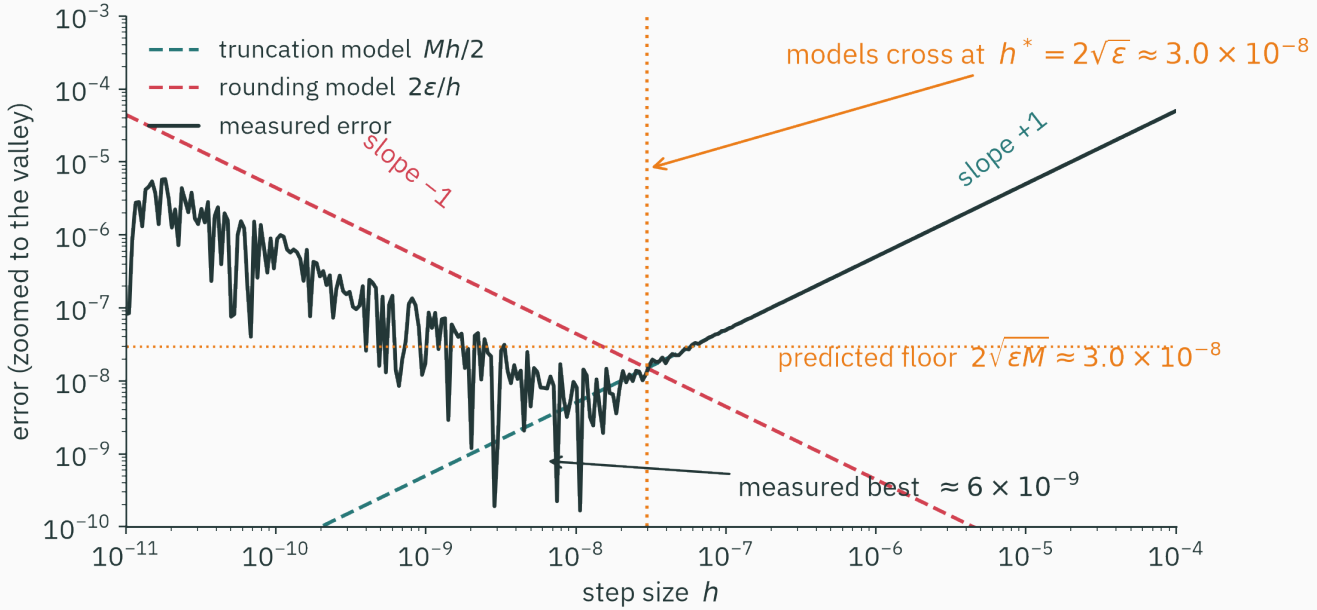
$h^* = 2\sqrt{\varepsilon/M} \approx \sqrt{\varepsilon}$, **best error** $\approx \sqrt{\varepsilon}$ **— for** $M \sim 1$**, both are “root machine epsilon”.**

Float64: $h^* = 2\sqrt{2.2 \times 10^{-16}} \approx 3 \times 10^{-8}$, error floor $\approx 3 \times 10^{-8}$.

⭐ Step 3: verify against the machine

D DERIVATION V VISUAL

Zoom into the U-curve's valley and lay the model on top of the measurement:



The terms cross near $h^* \approx 3 \times 10^{-8}$. The predicted worst-case floor is 3×10^{-8}, the measured best error is 6×10^{-9}, and the two asymptotic slopes are ± 1.

The precision floor: roughly half the digits

Relative error $\sqrt{\varepsilon}$ means: **of the digits your float carries, forward differences keep half.**

Format	ε (L2′s table)	digits carried	digits surviving $\sqrt{\varepsilon}$
float64	2.2×10^{-16}	~16	~8
float32	1.2×10^{-7}	~7	~3.5
float16	9.8×10^{-4}	~3	~1.5 — noise

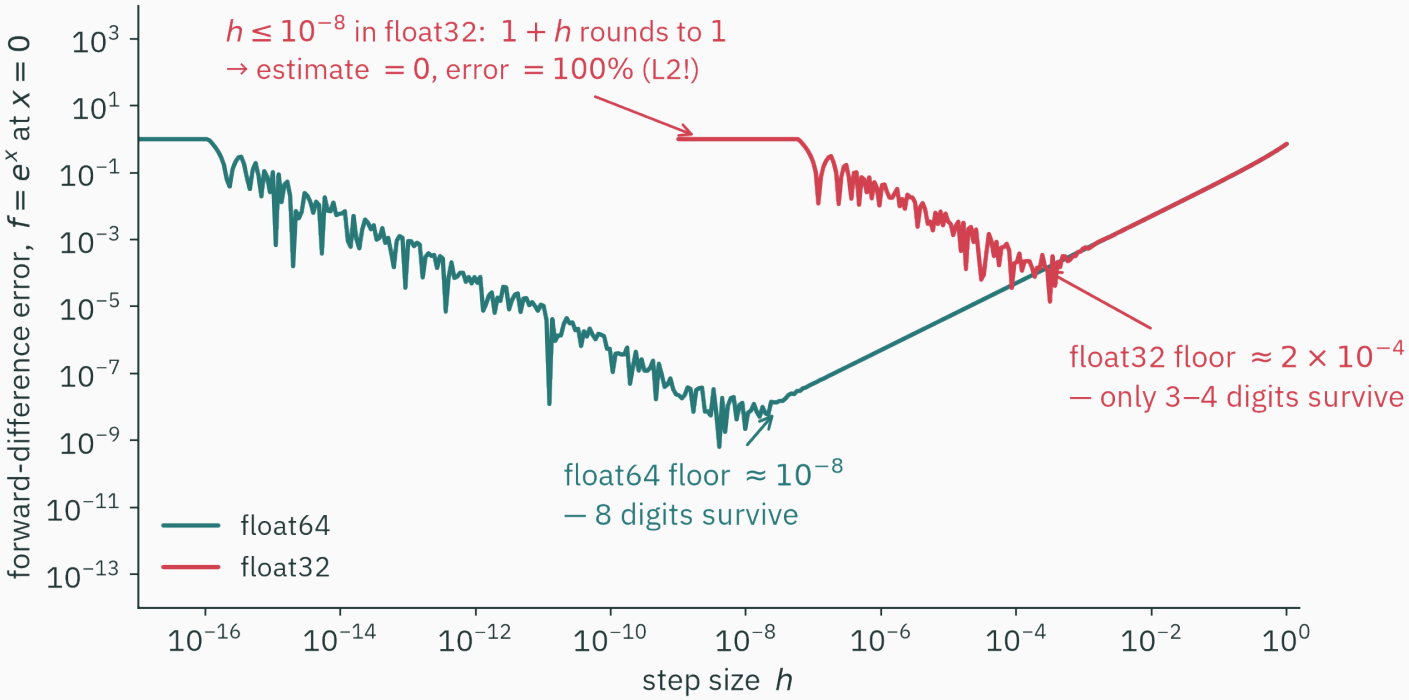
With default settings, a float32 numerical check may not distinguish a correct gradient from an error around 10^{-3}. Gradient-checking tools therefore recommend double precision and warn that lower-precision inputs may fail their default tolerances.

The precision floor: roughly half the digits

Rule of thumb worth memorizing: finite differences cost you **half your precision**. Exactness isn't on the menu — at any h.

Same experiment, half the bits

V VISUAL



Same f, same code — float32's floor sits roughly **ten thousand times higher**, and below $h = 10^{-8}$ its estimate is exactly 0 because `1.0 + 1e-8 == 1.0` in float32.

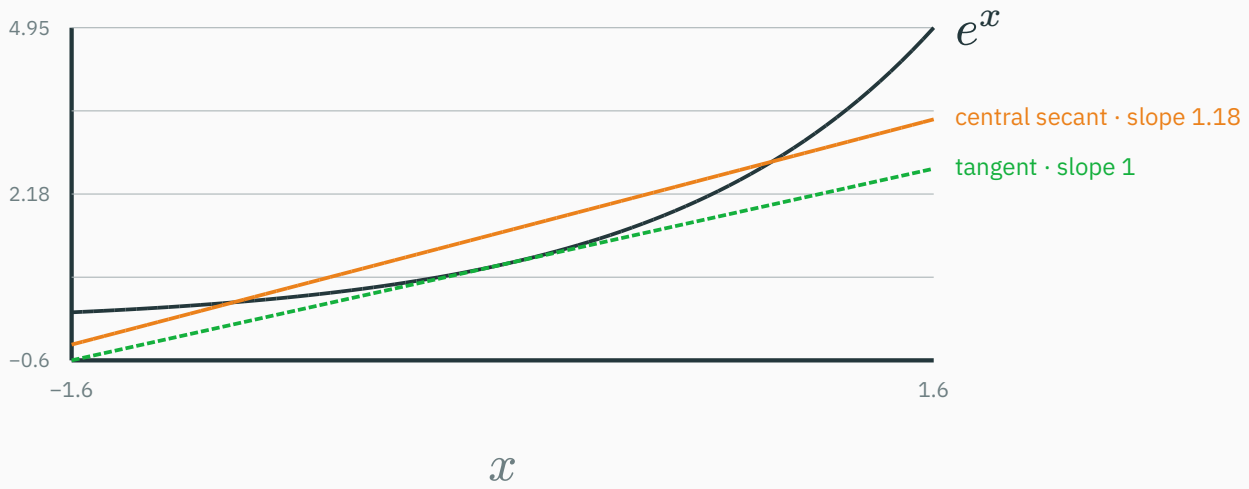
Central differences: a better secant

Symmetrize the secant

V VISUAL

Don’t lean on one side — straddle the point: sample at $x - h$ **and** $x + h$.

$$C_h f(x) := \frac{f(x+h) - f(x-h)}{2h}$$



Same generous $h = 1$: the one-sided secant said 1.72; the symmetric one says 1.18 — already near-parallel to the tangent.

Why symmetry buys a whole order

Write Taylor **both ways** and subtract — watch the even terms die:

$$f(x+h) = f + hf' + \frac{h^2}{2}f'' + \frac{h^3}{6}f''' + \dots \qquad f(x-h) = f - hf' + \frac{h^2}{2}f'' - \frac{h^3}{6}f''' + \dots$$

$$C_h f(x) = f'(x) + \underbrace{\frac{h^2}{6}f'''(\xi)}_{\text{truncation now } O(h^2)} \qquad \text{— the } f'' \text{ term cancelled } \textbf{exactly}$$

Halve h → **quarter** the truncation error. At $h = 10^{-2}$: forward is off by 5×10^{-3}, central by 1.7×10^{-5}.

Same two evaluations as forward (reused smartly), one order better.
This is why every serious gradient checker is central.

Central differences: the same tradeoff at lower error

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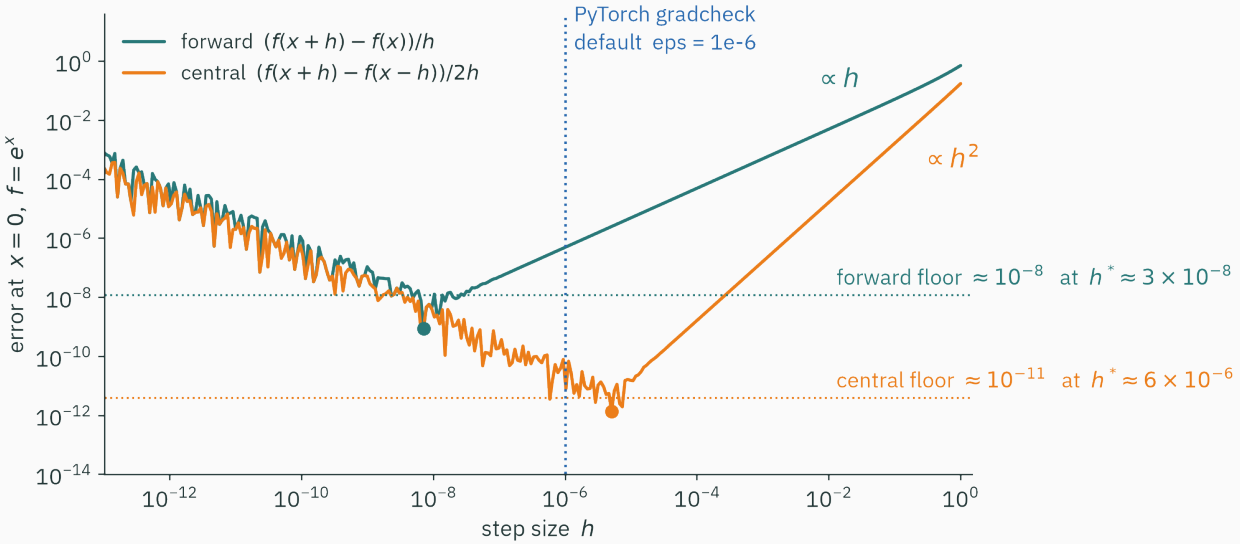


⭐ redo (practice): minimize $E(h) = Mh^2/6 + \varepsilon/h \to h^* \propto \varepsilon^{1/3} \approx 6\times10^{-6}$, floor $\propto \varepsilon^{2/3} \approx 10^{-11}$.

For float64 central differences, the balance between truncation and rounding error places the optimal h near 10^{-6}.

Checkpoint: choosing the step size

Q QUESTION

Central differences, float64, well-behaved f. You shrink h from 10^{-5} to 10^{-7} and the error gets worse. What happened?

A. Truncation error grew — h is still too big

B. You crossed h^*: rounding's ε/h term now dominates

C. Central differences are only valid for $h > 10^{-5}$

D. The function must not be differentiable at x

Answer: choosing the step size

A ANSWER

Correct: B — past h^*, rounding dominates

Why: For float64 central differences, $h^* \approx 6 \times 10^{-6}$. Below it, truncation continues to shrink as h^2, but cancellation error grows as ε/h. Therefore smaller h is not always more accurate.

The wall: one evaluation per input

A gradient is n separate questions

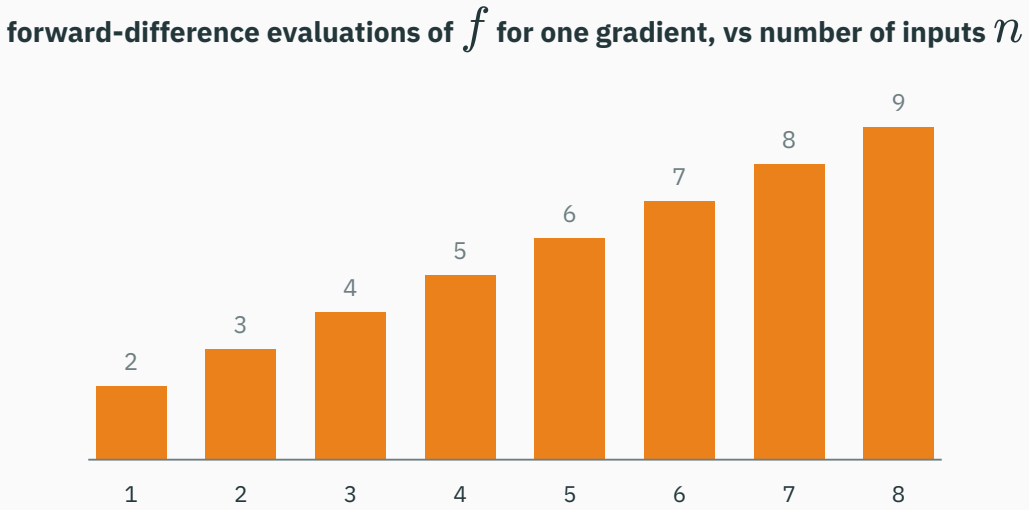
For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient wants $\partial f / \partial \theta_i$ for **every** i — and a nudge can only ask about **one input at a time**:

```
def grad_fd(f, theta, h=1e-6):
    g = np.zeros_like(theta)
    for i in range(len(theta)):          # one loop turn PER parameter
        e = np.zeros_like(theta); e[i] = h
        g[i] = (f(theta + e) - f(theta - e)) / (2 * h)
    return g                             # cost: 2n evaluations of f
```

Nudging θ_3 tells you **nothing** about θ_7. There is no bulk discount.

Function-evaluation cost grows with input dimension

V VISUAL



The staircase is the point: $n + 1$ evaluations (forward), $2n$ (central) — **linear in the number of parameters**, every single gradient step.

Now scale it to a real model

One forward pass of a big model ≈ 60 ms. One finite-difference gradient:

Model	**Parameters n**	**Time for ONE gradient**
linear regression	10	~0.7 s — fine
small MLP	10^5	1.7 hours
GPT-3	1.75×10^{11}	**~330 years**

Training needs **millions** of gradient steps. Finite differences don't need a better h — they need a different job.

Finite differences are suitable for occasional checks on small tensors, but not for every training step of a large model.

Comparison after two approaches

Route	**Accuracy**	**Cost for ∇f, $f : \mathbb{R}^n \to \mathbb{R}$**
symbolic	exact	expression swell; needs a formula
finite differences	$\approx$, floor $\sqrt{\varepsilon}$ — half your digits	$n + 1$ evaluations
wanted	exact	~cost of running f, no formula

The remaining goal is a formula-free derivative at roughly the cost of evaluating f. Dual numbers provide the construction.

Dual numbers: an infinitesimal you can compute with

Remove the quadratic term algebraically

Stare at why truncation existed at all:

$$f(x+h) = f(x) + hf'(x) + \underbrace{\frac{h^2}{2}f'' + \ldots}_{\text{everything we DON'T want}}$$

The unwanted terms all carry h^2 or higher. Finite differences make h^2 *small* and pay $\sqrt{\varepsilon}$ for it.

We introduce a formal unit with $\varepsilon^2 = 0$ but $\varepsilon \neq 0$. No real number has this property; it defines a different algebra.

Dual numbers extend real arithmetic with a formal unit ε satisfying $\varepsilon^2 = 0$.

Meet the dual numbers: decree $\varepsilon^2 = 0$

A **dual number** is $a + b\varepsilon$ — carry the pair (a, b), compute with ordinary algebra plus one new rule, $\varepsilon^2 = 0$:

Operation	**Result — just expand, then apply $\varepsilon^2 = 0$**
$(a + b\varepsilon) \pm (c + d\varepsilon)$	$(a \pm c) + (b \pm d)\varepsilon$
$(a + b\varepsilon)(c + d\varepsilon)$	$ac + (ad + bc)\varepsilon + \underbrace{bd\varepsilon^2}_{=0} = ac + (ad + bc)\varepsilon$

Look hard at the product's ε-part: $ad + bc$ — if b, d were "derivatives of a, c"... that is the **product rule**, appearing uninvited.

Two epsilons in this lecture, by historical accident: **machine epsilon** ε_{64} (a float's gap, L2) and this **dual unit** ε (an algebraic symbol). Unrelated beasts. From here on, ε means the dual unit.

Example: $(3+\varepsilon)^2$

Feed $x = 3 + \varepsilon$ — "3, plus a formal nudge" — through plain squaring:

$$(3+\varepsilon)^2 = 9 + 6\varepsilon + \varepsilon^2 = 9 + 6\varepsilon$$

Read the pair: value $9 = 3^2$ ✓ and ε-coefficient $6 = 2 \cdot 3 = \left(\frac{\mathrm{d}}{\mathrm{d}x}\right) x^2 \,|_{x=3}$ ✓

Once more: $(3+\varepsilon)^3 = 27 + 27\varepsilon$ — and indeed $\left(x^3\right)' = 3x^2 = 27$ at $x = 3$. ✓

Push $x + \varepsilon$ through the arithmetic and read the derivative from the ε-coefficient. There is no finite step or subtraction of nearby values.

Why the ε-slot always holds the derivative

Not a coincidence — it's Taylor, with the tail **annihilated by algebra**. For smooth f, substitute a dual $t = b\varepsilon$ into the expansion:

$$f(a + b\varepsilon) = f(a) + f'(a)b\varepsilon + f''\frac{a}{2}b^2\underbrace{\varepsilon^2}_{0} + \ldots = f(a) + f'(a)\, b\, \varepsilon$$

Every term we fought as "truncation error" contains ε^2 — and ε^2 **is zero**. Nothing is truncated; nothing is left to truncate.

Finite differences make the Taylor tail small and pay $\sqrt{\varepsilon_{64}}$. Dual numbers make it zero — by decree.

Thus the finite-difference error sources are absent: there is no truncated finite step and no subtraction of nearby values.

Every primitive learns one dual rule

$f(a + b\varepsilon) = f(a) + f'(a)b\varepsilon$ hands each primitive its own one-liner:

Primitive	**Dual rule**	**...which is**
$u + v$	$(a + c) + (b + d)\varepsilon$	sum rule
$u \cdot v$	$ac + (ad + bc)\varepsilon$	product rule
e^u	$e^a + e^a b\varepsilon$	chain rule for $\exp$
$\sin u$	$\sin a + (\cos a)b\varepsilon$	chain rule for $\sin$
$1/u$	$1/a - (b/a^2)\varepsilon$	reciprocal rule (practice)

A dozen rules cover a numerical library. In code this is **operator overloading**: teach $+$ and $*$ to a new class, and every program built from them differentiates itself.

A `Dual` class in ten lines

```
class Dual:
    def __init__(self, v, d):
        self.v, self.d = v, d                  # value, derivative
    def __add__(self, o):
        return Dual(self.v + o.v, self.d + o.d)
    def __mul__(self, o):
        return Dual(self.v * o.v,
                    self.v * o.d + self.d * o.v)   # product rule

x = Dual(3.0, 1.0)          # x = 3 + ε   (derivative of x wrt x is 1)
print((x * x).d)            # 6.0   — exact d/dx x² at 3
print((x * x * x).d)        # 27.0  — exact d/dx x³ at 3
```

Every $*$ applies the product rule to the tangent component. No step size is present in this code.

Forward mode: derivatives that ride along

The same recipe, now carrying two columns

Forward-mode AD = evaluate f step by step, dual all the way. Take $f(x, y) = (x + y) \cdot x^2$ at $(3, 1)$ — for $\partial f/\partial x$, **seed** $\dot{x} = 1, \dot{y} = 0$:

Step	**Value**	**Tangent rule**	**Tangent**
x (leaf)	3	seed	$\dot{x} = \mathbf{1}$
y (leaf)	1	seed	$\dot{y} = 0$
$a = x + y$	4	$\dot{a} = \dot{x} + \dot{y}$	1
$b = x^2$	9	$\dot{b} = 2x\dot{x}$	6
$f = a \cdot b$	36	$\dot{f} = \dot{a}b + a\dot{b}$	$9 + 24 = \mathbf{33}$

$f = 36$, $\partial f/\partial x = 33$ **— one sweep, carrying value and tangent.**

Tangents ride the same arrows

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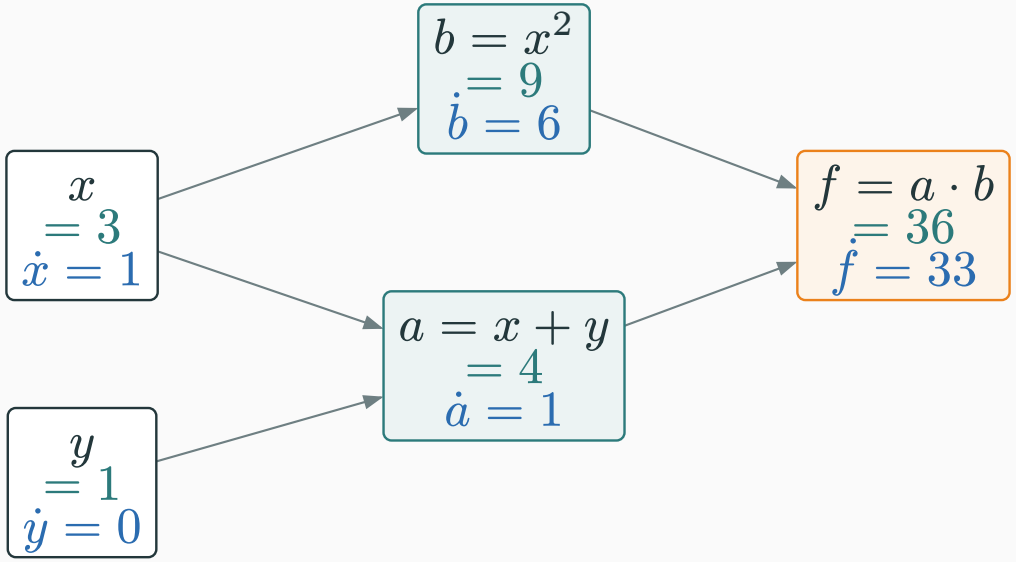


- Values (teal) flow left → right; tangents (blue) flow **with** them, through the **same** graph, in the **same** pass
- Hold this picture: L11 will send a different quantity through these arrows — **backward**

Where 33 came from: two routes, added

x reaches f two ways — through a and through b — and the tangent of the product added one contribution per route:

$$\dot{f} = \underbrace{\dot{a}b}_{\text{via } a:\,1\cdot 9=9} + \underbrace{a\dot{b}}_{\text{via } b:\,4\cdot 6=24} = 33$$

Sanity check by L8′s rules, on the closed form $f = x^3 + x^2 y$:

$$\frac{\partial f}{\partial x} = 3x^2 + 2xy = 27 + 6 = 33 \quad \checkmark$$

File away “**sum over routes**” — it is the multivariate chain rule wearing graph clothes, and it is the exact spot where L11′s backward pass will plug in.

The second input needs a second pass

$\dot{f}$ answered **one** question: sensitivity to x. For $\partial f/\partial y$, re-run with the other one-hot seed $\dot{x} = 0, \dot{y} = 1$:

Step	Value	Tangent (new seed)
$a = x + y$	4	$\dot{a} = 0 + 1 = 1$
$b = x^2$	9	$\dot{b} = 2 \cdot 3 \cdot 0 = 0$
$f = a \cdot b$	36	$\dot{f} = 1 \cdot 9 + 4 \cdot 0 = \mathbf{9}$

$\nabla f(3, 1) = (33, 9)$ **in two passes: one seed direction per input.**

Keep $(33, 9)$ warm — L11 re-derives this exact pair by a very different route.

The code agrees, digit for digit

The ten-line `Dual` class, on the two-variable f — same seeds, same numbers:

```
def f(x, y):
    return (x + y) * x * x      # ordinary code — no rewrite needed

print(f(Dual(3.0, 1.0), Dual(1.0, 0.0)).d)   # 33.0   ∂f/∂x  (seed x)
print(f(Dual(3.0, 0.0), Dual(1.0, 1.0)).d)   # 9.0    ∂f/∂y  (seed y)
```

No `h` anywhere in this program — so nothing to tune, no U-curve, no floor. The trichotomy's third road is **real**.

For comparison, central differences at a near-optimal h return approximately 32.99999999. `Dual` returns the floating-point result of the derivative arithmetic, nearest to 33 in this example.

Forward mode in the wild

Real frameworks ship exactly this trick (as “jvp” — more in a moment):

```
import jax
f = lambda x, y: (x + y) * x**2
val, dfdx = jax.jvp(f, (3.0, 1.0), (1.0, 0.0))   # value 36.0, tangent 33.0
```

- The seed needn’t be one-hot: seeding (v_1, v_2) computes $v_1 \partial_x f + v_2 \partial_y f = \nabla f \cdot \boldsymbol{v}$ — a **directional derivative** (L8 callback!)
- For vector-valued f, that dot product becomes Jv (L9′s Jacobian): a **Jacobian–vector product**, JVP — computed without ever building J

Forward mode requires one pass per input direction

The finite-difference approximation is gone. Now count passes for the function shape used in ML:

Function shape	**Forward-mode passes**	**Preferred mode**
$f : \mathbb{R} \to \mathbb{R}^m$ (one knob, many outputs)	**1**	perfect
$f : \mathbb{R}^n \to \mathbb{R}$ (a **loss**: $n = 10^9$ knobs, one number)	n — one per seed	330 years again

One seed vector is propagated per pass. Computing every coordinate of a scalar-output gradient still requires n forward-mode passes.

Forward mode avoids finite-difference error, but a full gradient of $f : \mathbb{R}^n \to \mathbb{R}$ still needs one pass per input direction.

Checkpoint: dual arithmetic

Q QUESTION

In dual arithmetic, $(2+\varepsilon)^3$ evaluates to...

A. 8

B. $8+3\varepsilon$

C. $8+12\varepsilon$

D. $8+12\varepsilon+6\varepsilon^2$

Answer: dual arithmetic

A ANSWER

Correct: C — $8 + 12\varepsilon$

Why: $(2+\varepsilon)^2 = 4 + 4\varepsilon$; then $(4+4\varepsilon)(2+\varepsilon) = 8 + 4\varepsilon + 8\varepsilon = 8 + 12\varepsilon$. The ε-slot holds $12 = 3 \cdot 2^2$ — exactly $(x^3)'$ at $x = 2$. Option D forgot the decree: any ε^2 term is **already** zero, not "small".

From forward mode to reverse mode

`gradcheck`, re-read with today's eyes

```
gradcheck(my_op, (x,))    # float64 recommended; default eps=1e-6; central formula
```

Design choice	Which is exactly…
central, not forward	$O(h^2)$ truncation — one order better, same evaluations
float64 recommended	float32's finite-difference floor is often too high for the default tolerances
eps = 1e-6	central's float64 sweet spot $h^* \sim \varepsilon_{64}^{1/3}$ — the U-curve's flat bottom
tests, not training	$2n$ evaluations: affordable once on a tiny operation, too costly for each training step

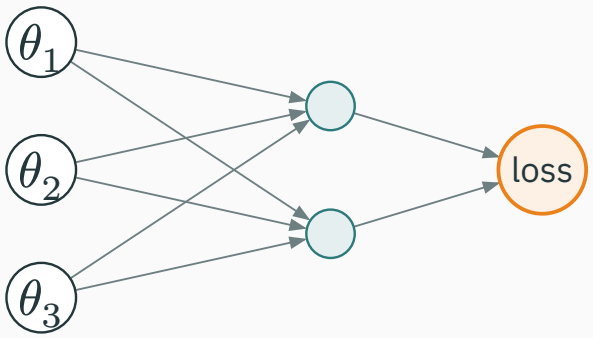
Gradient checking compares an analytical/autodiff gradient with an independent finite-difference approximation.

Method comparison

Method	**Assessment**	**Cost for ∇f, $f : \mathbb{R}^n \to \mathbb{R}$**
symbolic	exact, but expressions explode	—
finite differences	truncation vs rounding — both lose	n evals, **approximate**
forward-mode AD	no finite-difference approximation; carries (value, derivative) pairs	n **passes** — one per input
reverse-mode AD	L11	?

Forward mode is efficient for few-input, many-output functions. For a many-input, scalar loss, reverse mode has the favorable pass count.

Look at the shape of ML's function



$\vdots$ 10^9 of these

- Forward mode seeds an **input** → the wide end: 10^9 seeds, 10^9 passes
- But the narrow end has exactly **one** node — the loss...

Reverse mode seeds the scalar output and propagates adjoints backward through the graph, producing all input partial derivatives in one reverse sweep. L11 derives it.

Interactive: three roads on one screen

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

The **autograd** article puts symbolic, finite-difference and autodiff methods side by side on a live computation graph. Rebuild today's $f = (x + y) \cdot x^2$, vary h to see the U-curve, then switch to the graph walk and recover 33 and 9 without an h parameter.

Ten minutes before the next lecture: in the same article, press "reverse" on the graph walk and watch **which direction** the numbers travel. You'll have guessed most of L11.

Practice problems

Try on paper; verify in the T5 notebook (it pairs with L10–L11).

1. Estimate $\frac{\mathrm{d}}{\mathrm{d}x}\sin x$ at 0 with $h = 0.1$, forward and central. Both beat their advertised orders — which Taylor term vanished?
2. ⭐ redo for central: minimize $E(h) = Mh^2/6 + \varepsilon/h$; check h^* and floor against the figure.
3. Your GPU is float16 ($\varepsilon \approx 10^{-3}$). Can **any** h pass a 10^{-5}-tolerance gradient check? Use the $\sqrt{\varepsilon}$ rule.
4. Find $c + d\varepsilon$ with $(a + b\varepsilon)(c + d\varepsilon) = 1$; match your d to $(1/x)'$.
5. Trace forward mode on $g(x, y) = xy + x$ at $(2, 3)$, both seeds. Check: $\partial g/\partial x = y + 1$, $\partial g/\partial y = x$.
6. $n = 10^7$ parameters, 10 ms per pass: time one central-difference gradient, then one forward-mode gradient. (Both answers should make you want L11.)

⭐⭐⭐ Complex-step differentiation

★ OPTIONAL

Complex-step differentiation provides a related way to avoid subtractive cancellation:

$$f(x+ih) = f(x) + ihf'(x) - \frac{h^2}{2}f''(x) - i\frac{h^3}{6}f''' + \ldots \quad \Rightarrow \quad f'(x) = \frac{\operatorname{Im} f(x+ih)}{h} + O(h^2)$$

The imaginary part starts at **zero** — extracting it subtracts nothing. No cancellation → no left wall → h can be absurdly small:

```
>>> np.imag(np.exp(0 + 1e-200j)) / 1e-200
1.0                                  # exact — with h = 10⁻²⁰⁰ (!)
```

$i^2 = -1$ only leaks into the **real** part at $O(h^2)$ — harmless. Dual numbers go one step further: $\varepsilon^2 = 0$ kills the leak entirely. (Squire & Trapp, 1998 — beloved in engineering codes that predate autodiff.)

Lecture 10 — summary

- **Three approaches**: symbolic (may swell) · numerical (approximate) · automatic (chain rule on code).
- **Finite-difference error**: truncation $Mh/2$ (L7) vs rounding $2\varepsilon/h$ (L2) — the U-curve and its floor.
- ⭐ $h^* \approx \sqrt{\varepsilon}$, best error $\approx \sqrt{\varepsilon}$ — **half your digits**; central improves to $\varepsilon^{2/3}$. Hence `gradcheck`: float64 · central · `eps = 1e-6`.
- **The wall**: one nudge per input → $n+1$ evals per gradient — ~330 years/step for GPT-3.
- **Forward mode**: $\varepsilon^2 = 0$ → $f(a+b\varepsilon) = f(a) + f'(a)b\varepsilon$ for the lifted primitives; sweep (value, tangent) once per **input** seed — $\nabla f(3,1) = (33, 9)$ in two passes.

Lecture 10 — summary

Read before L11 — MML §5.6 (start); Baydin et al., *AD in ML: a Survey*, §2–3 (skim — today's trichotomy is their map). **T5** drills the dual trace, then L11's backward walk on the **same** graph.

Finite differences balance truncation against floating-point rounding.

Next: **Reverse Mode & Backprop** — reverse the arrows: one pass for **all** inputs.